

ABSTRACT

As face masks have become an irreplaceable object in our current lives, a challenge in recreating the faces of people have become very essential for the integrity of the security systems we have established in our society. Although a lot of work [1, 3, 9, 10, 11] has already been done in the field of in-painting and facial reconstruction using Deep Neural Networks, not much has been done for unmasking of faces, especially for replicating the emotions that the person has occluded with the mask. This project proposes a model that uses the emotion label as a condition for generation of the unmasked image called the RAC-UNET or Residual Attention Conditional UNET. The dataset used to train is the famous RAFDB dataset with digitally imposed masks. The report also highlights the performance of the model by comparing it with already existing networks that generate the images without an emotion label input.

1. INTRODUCTION

Face masks, although an integral part of our lives right now, has always been a challenge for our computer vision systems. Not just masks, any sort of occlusion is enough to render our security systems useless as they are not equipped with technology to recognise and detect such individuals. This problem started peaking when people were forced to wear masks during the pandemic.

All the work that had been done previously had been focused only on blindly in-painting the individual's face without paying attention to the emotion of the person, while using other semantic features of the face like landmark points. However, for visually realistic predictions that are very close to the ground truth, we need to consider all the information available to the system including the emotion conveyed by the periocular region that can be read and given as an input to the in-painting network. Although it is difficult to predict the emotions conveyed by the periocular region as shown in previous works [1].

Then, a literature survey showed the possible ways to attempt the problem of Masked Face Inpainting using UNETs [2] which were initially made for biomedical segmentation and Residual Attention UNET (RA-UNET) [3] by Md. Hosen which utilized

Attention Gating [4] to help the model understand where it must pay attention for in-painting and Residual blocks [5] that helped improve performance by eliminating issues like Vanishing Gradients.

Also, training such models on small datasets are not useful as the model might then be biased and learn how to in-paint only one class of people.

So, the choice of the dataset was a crucial decision to make. The dataset RAFDB [6] (Real-world Affective Faces Database) was chosen as it contained images of faces of people of all age groups and genders. Then, a synthetical mask was imposed on the face digitally. The emotion labels for the faces were also provided by the authors. The emotions were classified into 7 classes as follows: surprise, fear, disgust, happiness, sadness, anger and neutral.

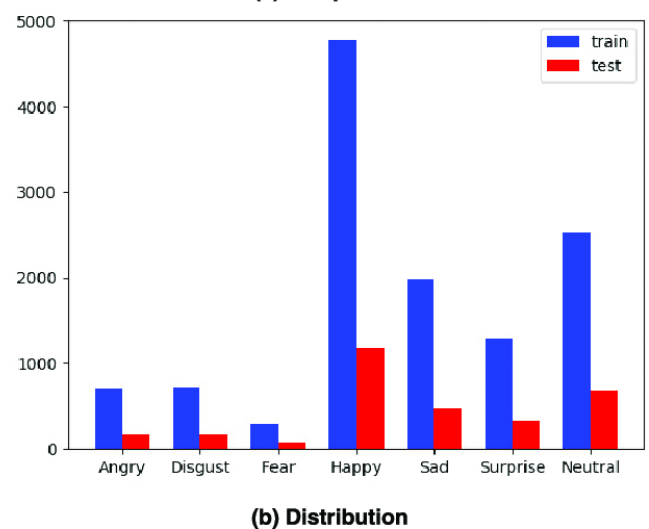
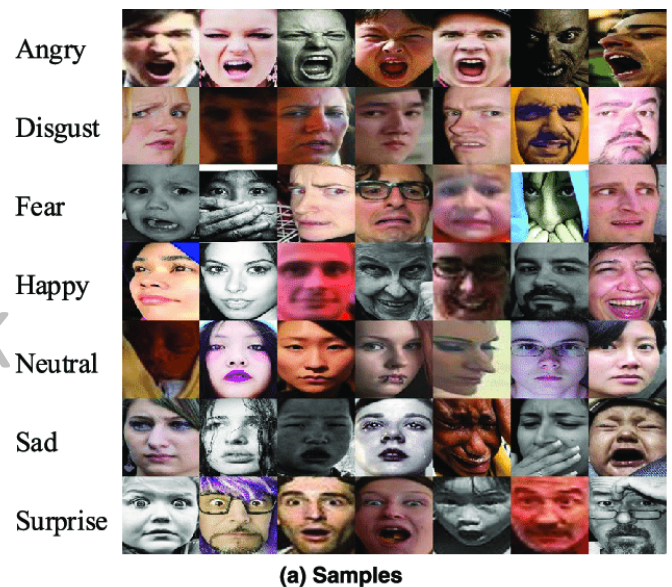


Fig 1: The RAFDB dataset samples with distribution of the emotions

After implementing these models and a bit of experimentation with the layers and residual blocks,

the RAC-UNET was developed. The RAC-UNET, as its name suggests, uses custom Residual blocks for encoding and decoding data and uses the Attention mechanism for skip connections that are passed by the previous layers while embedding the emotion label into the tensor representation of the input image. This method is successful in its endeavour of inpainting visually similar and striking images that has many times tricked my own classmates. Even though its results are nowhere near that of a Generative Adversarial Network (GANs) [7], which has long been the best algorithm for training image generation networks, this model is easy when it comes to training and can produce results surprisingly fast, in about 0.013 seconds per image on a Tesla P100 GPU which is good for processing of real-time videos (24 fps). Personally, this has been a really good introduction for me into the world of Data Science and Deep Learning, as these are currently the buzzword with the advent of ChatGPT and other Natural Language Processing and Diffusion models.

2. GROUND WORK

First, the necessary knowledge of coding in Pytorch [14] was acquired along with the fundamentals in Deep Learning [15] and the mathematics involved behind it from online learning platforms. Then, small models ranging from those classify points on the cartesian planes to actual image classifiers were trained to learn the workflow of a Deep Learning project in Pytorch. Following this, a comprehensive literature study was conducted to get an idea on how to tackle this problem. That is when I found the paper by Md. Hosen et al. which proposed a striking solution that was beginner friendly as well as performing. The next goal was to implement the paper which was not so easy given that my learning and courses only took me as far as developing simple models, not ones with skip connections, Residual layers, and Attention mechanisms.

So, an even simpler UNET with just Attention blocks was developed that could generate a face with some blurry patches. The network was trained with Structural Similarity Index Metric (SSIM) as used by Md. Hosen et al. in their paper along with Adam optimizer with learning rate = 0.001. The knowledge about SSIM and other criterion like Perceptual Loss and Peak Signal to Noise Ratio were also acquired. The implementation was initially done on Google Colab but due to convenience issues, I shifted to

Kaggle as they provided options like version control etc.

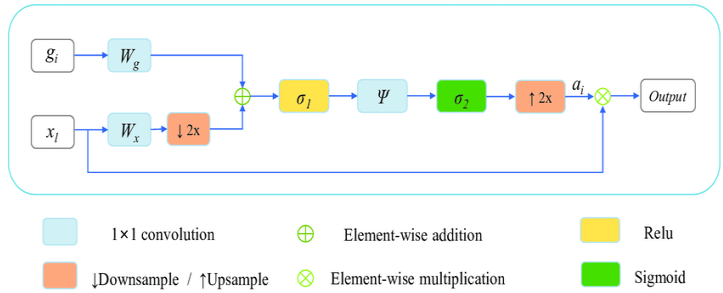


Fig 2: Attention Mechanism

The attention mechanism utilizes two signals or inputs, namely the Gating signal (input from the previous layer) and a Skip connection that is the output from a layer even further back in the pipeline. Both the inputs are passed through a 1×1 convolution and the skip connection is downsampled to the dimensions of the gating signal and then the two are added element-wise following which they are passed through an Activation function like ReLU or Sigmoid which provide non linearity to the models by selectively activating neurons depending on the input. Following this, the output is passed through another convolution and finally resampled before being multiplied element-wise with an identity skip connection.

The first few layers in the model learn low-level features like edges and outline of the faces etc. Hence, by using these skip connections, we are able to get back an accurate and structurally similar face while reconstructing the image. The next few layers learn high-level features like facial components etc. which are also crucial for reconstruction.

This then gave enough confidence and insight into replicating the Residual Attention U-NET (RA-UNET) which was still challenging as the topic of Residual Learning was still not very intuitive. Eventually, upon coding the RA-UNET, results were even sharper and had retained more facial features that were lost in the prediction of A-UNET. The network seemed to learn to in-paint emotions blindly as it learnt features during training. Training the RA-UNET took much more time as the Residual blocks allowed the network to learn more and hence required more epochs of training to converge the loss.

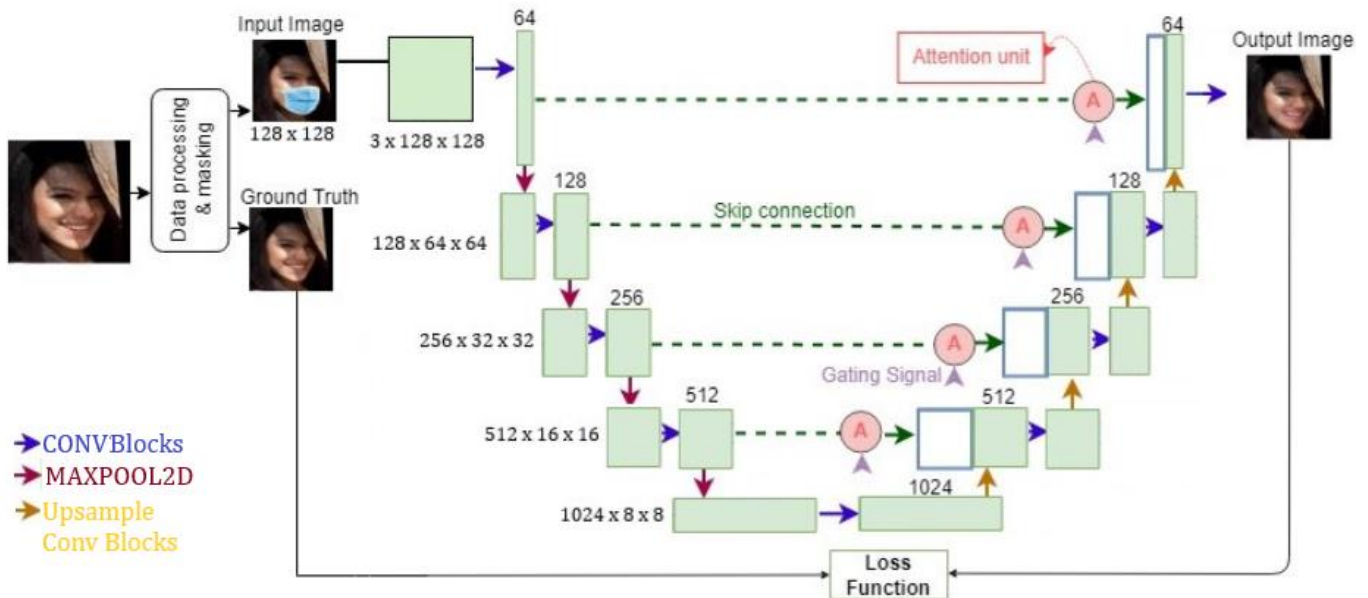


Fig 3: Attention UNET architecture

Above shown is the first model, Attention UNET that is a plain vanilla UNET with some attention mechanisms. Since it is a relatively simple architecture, the blurry results are justified. It is however able to generate back an image of the same input resolution.

Below shown (Fig 5) is the architecture proposed by Md. Hosen et al. [3] which uses Residual blocks (Fig 4). Residual blocks [5] are used to enhance the learning of networks by adding an identity of the input back to the results of convolutions. Since all these models work using the Gradient Descent algorithm given by Cauchy, a problem known as the Vanishing gradients arise if the gradients calculated during back propagation are very small, rendering the learning to almost zero as there is not enough value to move towards the minima efficiently. Residual networks resolve this because of the addition of identity to the output.

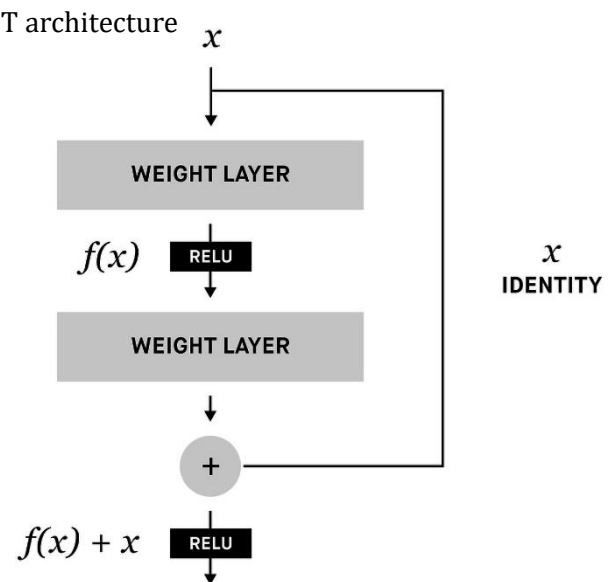
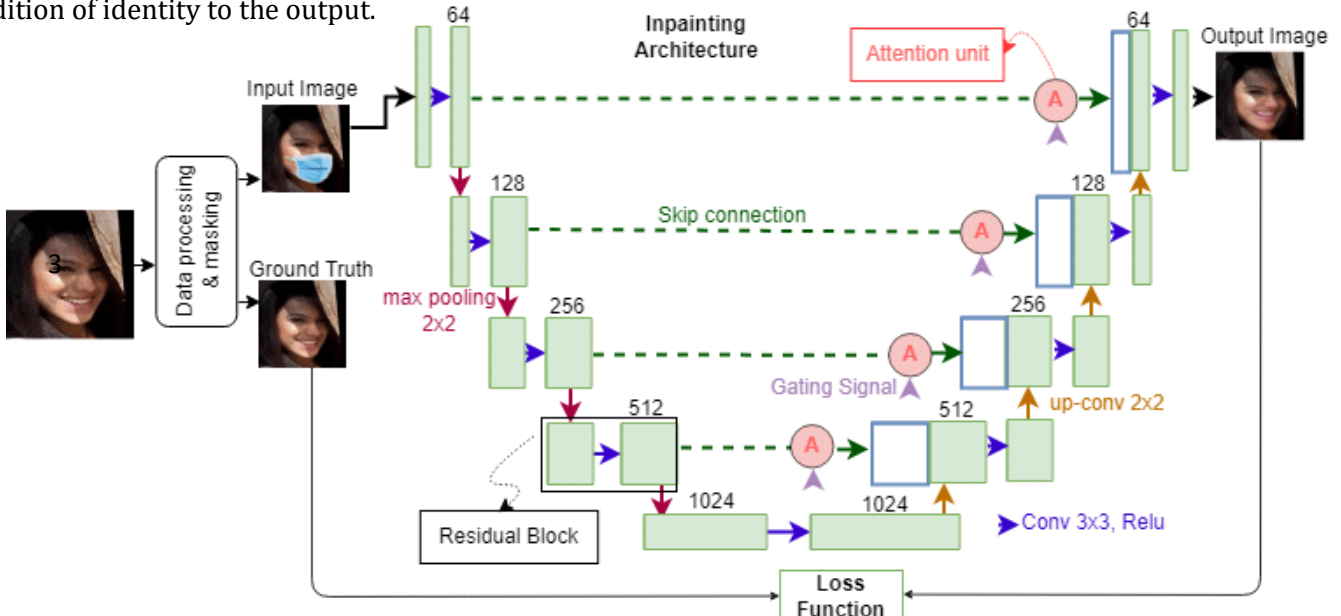


Fig 4 (above): Residual Blocks

Fig 5 (below): Residual Attention UNET architecture



But, the problem with these models as mentioned before is that it blindly learns the facial emotions from the unmasked region and tries to in-paint it which is not very efficient. Hence, a new model was developed that takes in the emotion label also as an input.

3. PROPOSED MODEL

The Residual Attention Conditional UNET uses embedding layers that take in the facial emotion labels as an input and concatenates the same with the tensor representation of the input image. This embedding layer biases the values slightly so that the model is able to learn how to predict the face for a given label. The concatenated tensor is then passed through an Up-feature Convolution layer that changes the no. of channels of the tensor back to 32 and then the process is similar to that of RA-UNET, but with some more modifications.

The proposed model, RAC-UNET uses custom residual layers for both, encoder layers and decoder layers so as to give the flexibility of choosing the number of layers and making the network deeper to learn more features and be able to accurately recreate the faces. The Upscaling layers of the decoder has inbuilt Attention gates to keep the model a bit more clean and easier to read.

These Residual Encoder blocks use 2D convolutional layers for down-sampling instead of MaxPool layers as it made sense for me to take a feature map instead of just taking the maximum of a 2×2 kernel which might lead to loss of data. This step however might slightly add to the complexity of the computations performed by the model and increase its size, which is a problem for another day.

The Decoding layers also use Transposed convolutions in the same manner instead of some regular Up-sampling with a bilinear function to just predict values based on the locality.

The training and testing setup on RAFDB is similar to the previous networks, using the same MS-SSIM loss function with ADAM optimizer. Another challenge that came up was that generic Dataset and Data Loader modules could not be used for this model as both the features and labels fed to the model are images. So, custom Dataset and Data Loader modules were also developed which could feed both the images and the emotion label to the model. One more instance of the same model was trained using VGG Perceptual Loss based on the studies presented in [8]. Models trained with VGG Perceptual loss also seems to perform better when it comes to dataset that incorporate facial emotions. However, this model took more time to converge and in order to maintain fair standards for comparison, all the models were trained for 250 epochs on the train set of RAFDB.

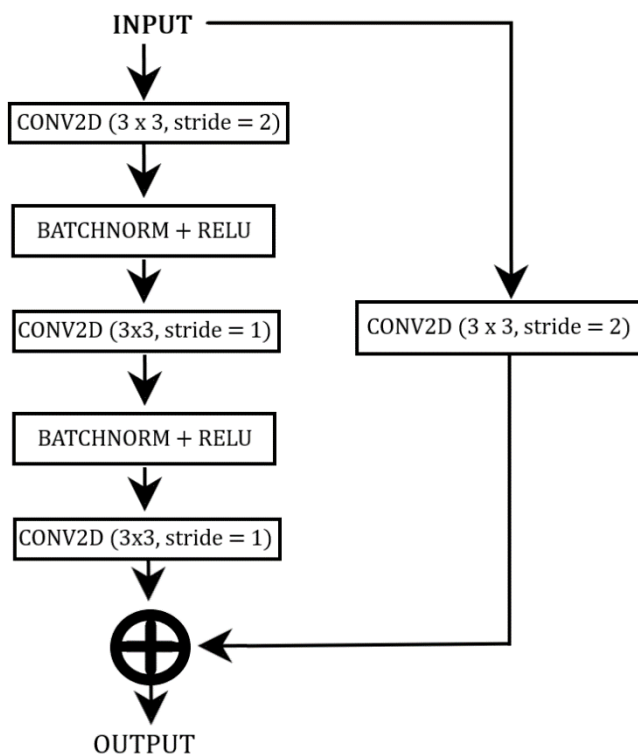


Fig 6: Custom Residual Block used for encoding in RAC-UNET

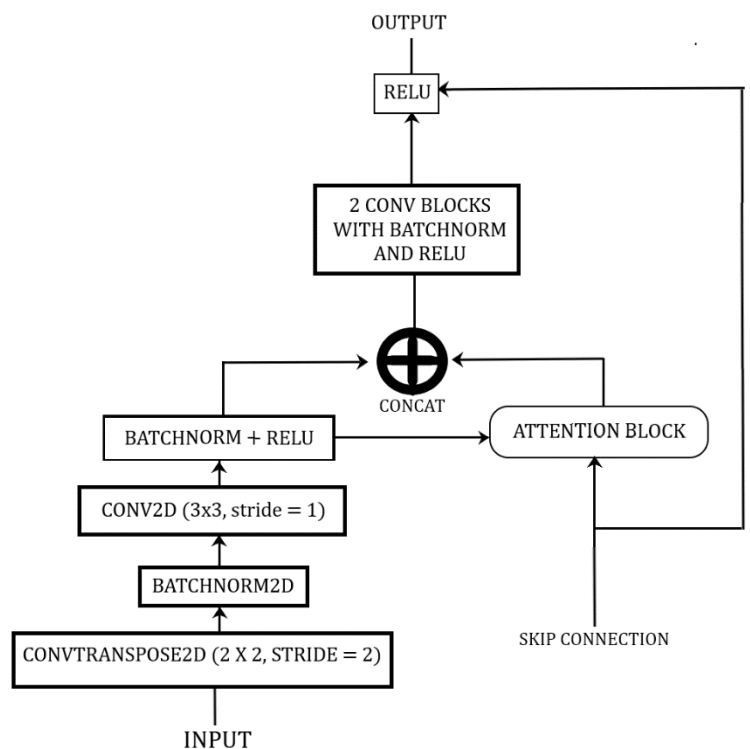


Fig 7: Custom Residual Block used for decoding in RAC-UNET

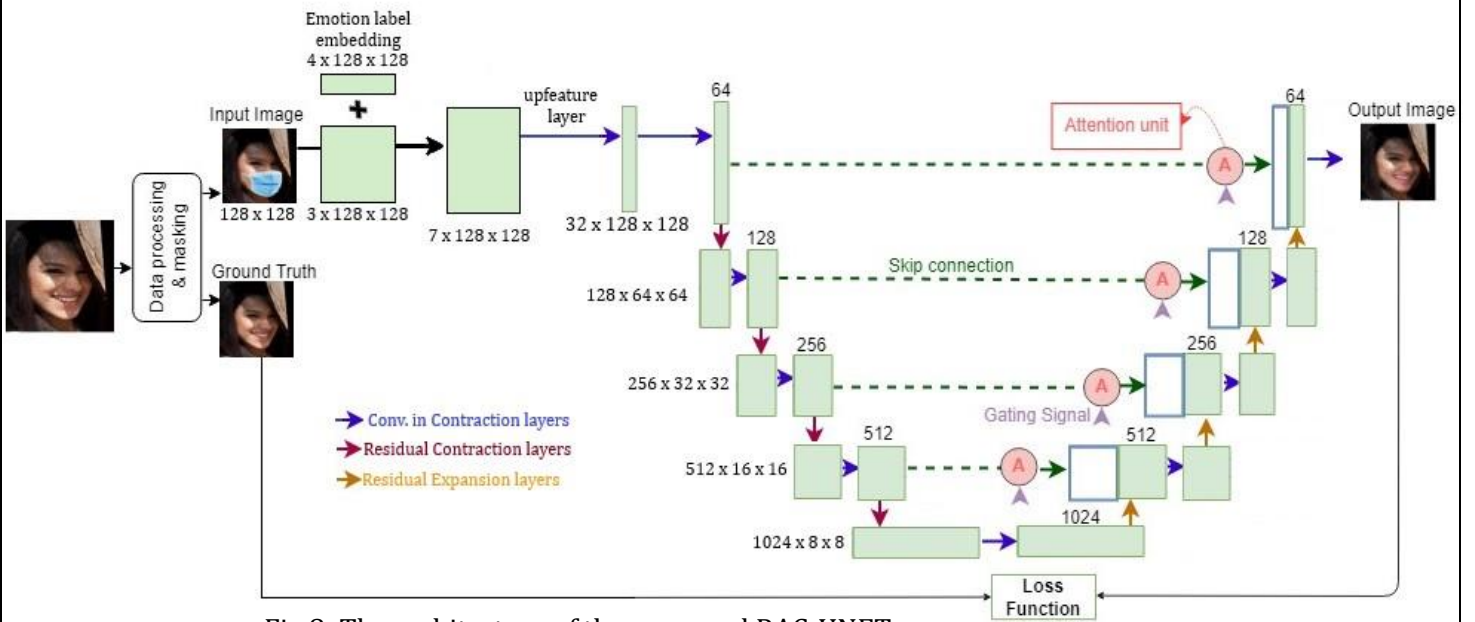


Fig 8: The architecture of the proposed RAC-UNET

4. EXPERIMENTS AND RESULTS

A series of experiments were conducted including testing the model with real masked images also. The results are as expected with the RAC-UNET beating the other two models. The metrics used for evaluation are Peak Signal To Noise Ratio (PSNR) which measures the power of the signal and the corrupting noise, Structural Similarity Index (MS-SSIM) which looks at a range of features like luminosity, similarity in the structure of images and VGG Perceptual Loss which shifts the given image by one pixel and compares it to the target image. PSNR as such is used for quantifying reconstruction quality of images from compressed images, which is not exactly suitable as a metric for our problem of inpainting but still we have considered it. Below is the table of results for each of the models.

The models were trained on Tesla P100 GPUs provided by Kaggle with batch size of 8. As mentioned earlier, all the models were trained for 250 epochs except A-UNET as it had converged before 250 epochs itself. The metric values for RAC-UNET with perceptual loss seem absurd as it takes a long time to train compared to the RAC-UNET trained on MS-SSIM, depicted in the 3rd column. Due to the lack of time, this problem could not be resolved. Clearly the RAC-UNET outperforms both A-UNET and the RA-UNET in terms of metrics.

Fig 10 shows a comparison of the images generated by all the 4 models, which also shows that the RAC-UNET is much better in terms of accurately recreating the image and maintaining the emotion feature of the face.

	A-UNET	RA-UNET	RAC-UNET	RAC-UNET (PERCEPTUAL)
PSNR	152.3708	120.8394	134.3380	60.5981
MS-SSIM	158.2562	265.9105	150.6435	704.7717
VGG Perceptual	15.8089	26.4389	11.8438	40.3354
Prediction Time	0.05184	0.05774	0.06536	0.06531

Fig 9: Performance Metrics (per batch of 8 images) of the various models in terms of Peak Signal To Noise Ratio (PSNR), Structural Similarity (SSIM), Perceptual loss and Prediction time

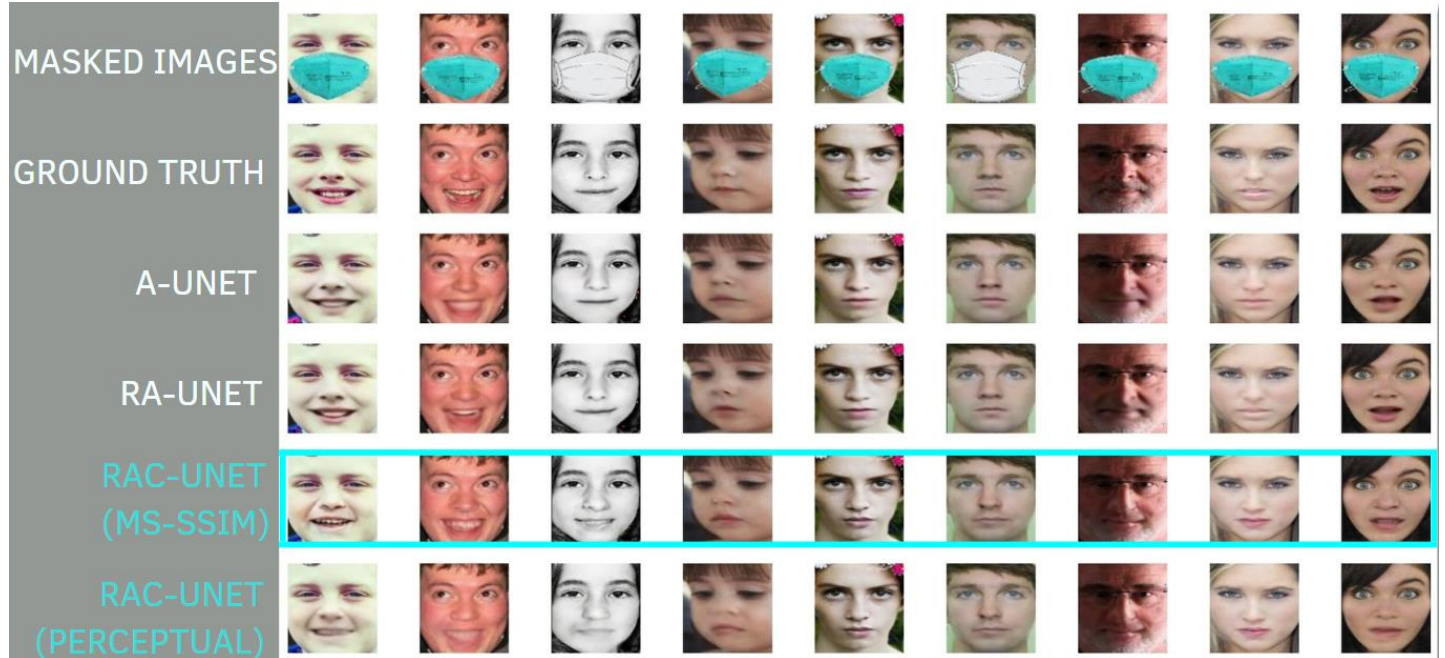


Fig 10: Comparison of the outputs of the four models - A-UNET, RA-UNET, RAC-UNET (MS-SSIM) and RAC-UNET (Perceptual Loss)

Although the metric values are not great for the RAC-UNET trained on Perceptual loss, its image recreation is on par with that of the MS-SSIM trained RAC-UNET. It is able to generate the images with some slight blurriness which will be eliminated on training and converging the model.

The models were also tested on the FER dataset provided by Sridhar Sola and Dr. Darshan Gera [1]. The results however were disfigured faces. Some instances of failure of the model are shown in Fig 11. These images turned out to be disfigured due to the occlusions which block important information required for the model to process the input and the other was a case of a badly positioned face, the kind of which is not plenty in the train dataset. The model also seems to be able to easily recreate images in which the faces show a happy or neutral emotion as the RAFDB dataset is composed majorly of such images. Data augmentation techniques can be used to increase the number of samples of the other emotions in order to prevent biases that arise due to training.

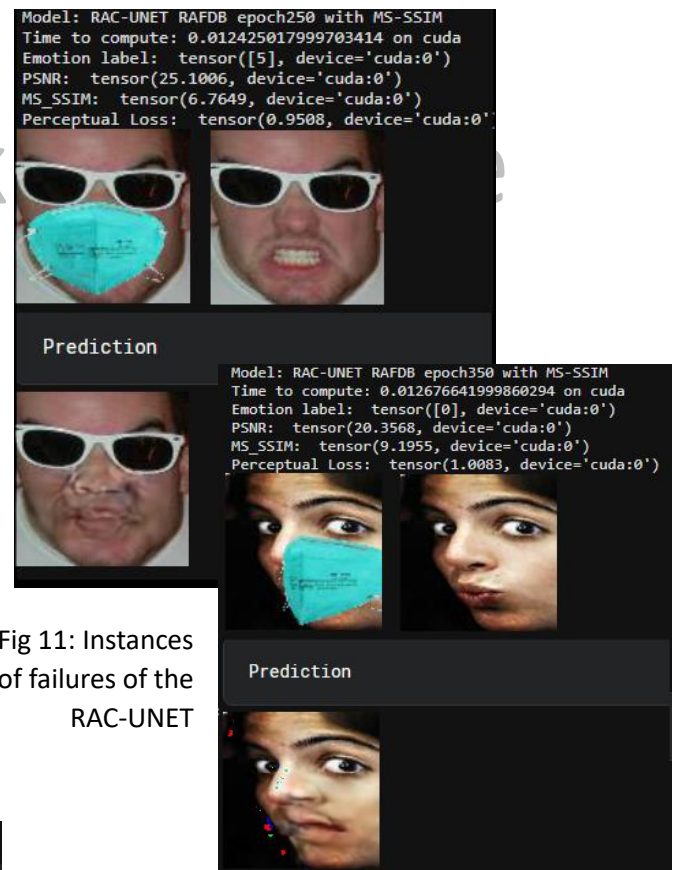


Fig 11: Instances of failures of the RAC-UNET



Fig 12: Example to show robustness in training

A good example of a prediction that shows the robustness of the training is shown in the figure 12. Sometimes, even when the image is bad, the model is able to predict reasonably well.

5. CONCLUSIONS AND FUTURE PLANS

5.1 CONCLUSIONS

As a part of the UMAT – 605 course, the basics of Artificial Intelligence, specifically Deep Learning and the development of Deep Neural Networks, Convolution Neural Networks and a bit of Generative models were studied thoroughly and the problem of Masked Face In-Painting with Emotion factor was attempted successfully and a custom model was developed from scratch (including coding of it) which was able to give really good results. The model was also evaluated based on different evaluation metrics and compared with other models, which it clearly outclasses by quite some margin. The prediction time of the model was also good in the neighbourhood of 0.015 seconds, making it a potential candidate for real-time video processing and unmasking.

However, when it comes to removing masks from the images that contained real masks unlike the digitally imposed masks, the model quite literally did not work, which brings me to the plans I have, to carry forward this exciting project that has kept me thrilled throughout this semester.

5.2 FUTURE WORK

- Try and train the model to work on images with real masks, unlike the masked RAFDB dataset used in this project. The work for this has already started.
- The current size of the model is about 256 MB which is too high. The next goal would be to reduce the model size and making it lightweight so that it can be run on embedded chips and other devices with low processing power.
- Enable the model to process real-time videos and give a video of people without masks live from some video feed like that of CCTVs.
- Solve the problem by implementing the current state-of-the-art techniques like GANs or Diffusion models, which can give insanely good image generations.
- Venture into other fields of AI like Audio processing, Natural Language Processing which is the current talk of the town with the advent of ChatGPT and other transformer models.
- Learn other platforms like Tensorflow and Keras in order to equip myself with other languages and not be bound by the limitations of a particular language. This was really felt when I tried to use TPUs on Kaggle in order to accelerate my model's training only to find that the support for TPU on Pytorch was broken.

6. References

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